

Foundations of NLP

Training and Evaluating a Small Transformer Language Model

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Bianca Schmid (Matr.-Nr.: [REDACTED]), Louisa [REDACTED] (Matr.-Nr.: [REDACTED]), Muriel Danilow (Matr.-Nr.: [REDACTED]),
Patricia [REDACTED] (Matr.-Nr.: [REDACTED])

Introduction

This notebook presents the training and evaluation of a small character-level Transformer language model using nanoGPT on the Tiny Shakespeare dataset. The goal is to compare a baseline model with multiple controlled hyperparameter experiments and to analyze their effect on optimization dynamics (training stability and convergence), validation performance, and qualitative text generation. All experiments vary only one hyperparameter at a time relative to the baseline, following the systematic methodology required in the assignment.

Setup and dataset

The nanoGPT repository was used as provided in the assignment. The setup, including repository cloning and package installation, was executed within the notebook environment. An automatic device selection was implemented to improve training efficiency, using a GPU (cuda) when available and otherwise falling back to CPU. In our case, training was performed on GPU in Colab for faster execution.

The Tiny Shakespeare dataset was used as specified in the assignment. Preprocessing was performed using the provided prepare.py script, which converts the text into a character-level representation and creates binary training and validation files (train.bin and val.bin) as well as metadata for the character vocabulary. The dataset is relatively small and highly stylized, which makes it suitable for analyzing the effect of hyperparameters on both convergence and generated text quality. Due to its small size and repetitive structure, the dataset is prone to overfitting, making it particularly suitable for analyzing generalization behavior and the effects of hyperparameters in small scale language models.

To facilitate structured experimentation, separate directories for configurations, logs, plots, and generated samples were created, enabling consistent experiment tracking and analysis.

Reproducibility

To ensure reproducibility, a global random seed (1337) was fixed across Python, NumPy, and PyTorch through a helper function. This ensures consistent initialization and reduces randomness between runs. However, exact reproducibility cannot be fully guaranteed on GPU due to potential nondeterministic behavior in underlying operations.

Overview of experiments

In addition to the baseline, we ran five controlled experiments. Each experiment changes exactly one hyperparameter while keeping all others fixed, as required in the assignment.

This allows the effect of each change to be interpreted clearly.
The experiments are defined as follows:

- **lr1e-4**: lower learning rate, `learning_rate = 1e-4` instead of `1e-3`
- **lr5e-3**: higher learning rate, `learning_rate = 5e-3` instead of `1e-3`
- **layer4**: shallower model, `n_layer = 4` instead of `6`
- **block128**: shorter context length, `block_size = 128` instead of `256`
- **dropout01**: lower dropout, `dropout = 0.1` instead of `0.2`

This selection allows us to study optimization (learning rate), model capacity (layers), context modeling (block size), and regularization (dropout).

All experiments were defined programmatically in a centralized structure, ensuring that only the intended hyperparameter was modified in each run and reducing manual errors.

To execute the experiments, a shared pipeline was implemented using helper functions. These functions automated configuration generation, training, logging, and sample generation, ensuring consistent and reproducible execution across all runs.

Quantitative analysis

Quantitative results are summarized in Table [A.1.3](#). Training and validation losses were extracted from the final evaluation step (step 5000), and wall-clock training time was recorded for each experiment. Since all runs use the same dataset, number of iterations, and evaluation protocol, the results are directly comparable. Results were generated automatically by parsing training logs and compiling them into a structured summary table, improving reproducibility and reducing manual errors. Validation loss is used as the primary metric for model quality. The results show a clear ranking across experiments. The best validation performance is achieved by `block_128` (val loss 1.4986), followed by `lr_1e4` (1.5090), `layer_4` (1.5268), and `lr_5e3` (1.5421). The baseline model performs worse (1.7103), while the weakest result is `dropout_01`, whose validation loss increases to 2.2670 despite having the lowest training loss (0.2529), indicating severe overfitting. The differences in generalization are clearly visible in the loss curves (see Appendix [A.1.2](#) and [A.1.1](#)). In the training loss ([A.1.2](#)), all models decrease rapidly at the beginning, but separate afterwards: `dropout_01` continues to decrease steadily and reaches the lowest training loss, while `lr_1e4` converges more slowly and remains at a higher level. In the validation loss ([A.1.1](#)), `dropout_01` reaches its minimum early (around 1000–1500 steps) and then increases continuously, while `block_128` and `layer_4` initially follow similar trajectories as the baseline but achieve lower validation loss in later stages and remain more stable. The baseline already exhibits moderate overfitting, with a training loss of 0.62 compared to a validation loss of 1.71. This effect becomes much more severe for `dropout_01`, where the model fits the training data extremely well but generalizes poorly, producing the largest gap across experiments. Reducing the context length, lowering the learning rate, or reducing model depth effectively reduces model capacity and prevents overfitting on this small dataset. Training efficiency further differentiates the models. `block_128` is not only the best-performing model in terms of validation loss, but also the fastest (~3.83 minutes), as halving the context length reduces computational cost. `layer_4` also improves efficiency

(~5.31 minutes) by decreasing model depth, while baseline and learning rate variations require similar training time (~7.9 minutes).

Perplexity values follow the same ranking as validation loss (e.g., block_128: 4.48 vs. dropout_01: 9.65), confirming the consistency of the results across metrics. On Tiny Shakespeare, the best validation performance comes from a shorter context window, while reducing dropout (weaker regularization) leads to severe overfitting. Since each experiment changes only a single hyperparameter relative to the baseline, these differences can be directly attributed to the respective design choices. This confirms that on small datasets, reducing model capacity or effective context can improve generalization by limiting overfitting. Overall, the results suggest that validation loss is a more reliable indicator of model quality than training loss.

Qualitative analysis

All models capture key aspects of the surface structure of Shakespearean drama in the generated samples (see [A.3](#)), including archaic language patterns (e.g., “thy”, “thou”), uppercase speaker names, turn-taking dialogue, and line breaks. Plausible character labels such as CORIOLANUS, MENENIUS, LUCIO, and KING EDWARD IV appear across all configurations. However, this structure is not maintained consistently: within each sample, there are also stretches of text without speaker labels or clear dialogue attribution. This indicates that the models have learned general formatting patterns, but do not reliably sustain them over longer sequences.

The baseline samples (see [A.3.1](#)) are clearly Shakespeare-like in tone and structure, but they break down semantically after short stretches. They contain invented or malformed words such as “baredilation,” “starried,” and “coverby,” and the syntax is often unstable. While individual lines can appear plausible, the dialogue quickly loses coherence across multiple lines. This suggests that the model captures local syntax and stylistic patterns better than long-range semantic coherence. This aligns with the baseline’s moderate validation loss and noticeable train–validation gap.

The lr1e-4 model shows slightly more controlled phrasing in some parts compared to the baseline, although structural inconsistencies remain frequent (see [A.3.5](#)). Malformed words such as “Redicle” and “accessation” still occur, and coherence is limited, but some passages appear less abrupt in their breakdown. This is consistent with its improved validation loss and indicates somewhat better generalization at a lower learning rate.

In contrast, the lr5e-3 model appears more unstable (see [A.3.6](#)). Some samples begin without speaker labels, and the structure fluctuates throughout. Sentences are often repetitive or awkwardly constructed, and coherence varies strongly between segments. While dialogue elements are still present, they are less reliably maintained, which reflects the weaker validation performance of this configuration. The layer4 model, despite having fewer parameters, produces relatively clean local phrasing in several parts (see [A.3.4](#)). As with other models, speaker labels and dialogue structure are present but not consistently sustained. The text still contains occasional malformed words such as “qoon,” and coherence remains limited, but transitions between lines are sometimes less abrupt than in the baseline. This is consistent with the quantitative finding that the smaller model generalizes better on this dataset.

The block128 model produces the most coherent outputs overall (see [A.3.2](#)). It frequently generates well-formed dialogue segments with plausible structure and more stable wording

across multiple lines. However, even here the formatting is not perfectly consistent, and segments without speaker labels still occur. The samples include invented words such as “glassion” and “dissonmenting,” but the overall flow is more controlled and breaks down less abruptly than in other models. This aligns with its best validation loss and suggests that, for this dataset, reducing context length can improve generalization without harming qualitative performance. The clearest signs of overfitting appear in the dropout01 model (see [A.3.3](#)). Although it reproduces dialogue formatting and speaker labels in parts, the text is highly unstable and contains many malformed words such as “compounteous,” “exervices,” and “excelleiw.” Sentences frequently lose grammatical structure, and coherence deteriorates quickly. This is consistent with its extremely low training loss combined with the worst validation loss, indicating that the model overfits the training data and fails to generalize. Clear signs of severe underfitting (e.g., completely incoherent or gibberish output) are not strongly observed in these experiments. Although unstable text, malformed words, and low coherence appear across models, these patterns alone are not sufficient to distinguish underfitting from overfitting. The difference becomes clear only when considering training and validation loss. Overall, the qualitative results closely reflect the quantitative findings. Models that achieve lower validation loss, such as block128 and layer4, produce more stable and coherent text, while configurations that reduce regularization or destabilize training lead to visibly worse outputs. These results confirm that lower validation loss correlates with more stable and coherent text generation, while overfitting manifests in visibly degraded outputs. This reflects a trade off between optimization speed and generalization.

4.4 Analysis questions

1. What happens when the learning rate is too high? Too low? How does this manifest in the loss curve and the generated text?

A learning rate that is too high tends to make optimization unstable because the updates overshoot good parameter regions, which usually appears as noisy or oscillating loss curves and weaker generated text. A learning rate that is too low makes learning slower, so the model may still improve but not reach the best possible quality within a fixed training budget. Comparing all three learning rates makes the trade-off clearer. The baseline ($1e-3$) achieves the lowest training loss (0.6200) but a relatively high validation loss (1.7103), indicating some overfitting (see [Table A.1.3](#)). The lower learning rate ($1e-4$) leads to a higher training loss (1.1538) but improves validation performance (1.5090), suggesting better generalization. The higher learning rate ($5e-3$) achieves a moderate training loss (0.8520) but does not improve validation loss (1.5421), indicating less efficient optimization.

Overall, this shows that at this scale, reducing the learning rate improves generalization even if optimization becomes slower, while increasing it harms stability without providing benefits.

2. What is the relationship between model size (layers/embedding) and validation loss? Is bigger always better at this scale and dataset size?

Our results show that increasing model size does not necessarily reduce validation loss. Reducing the number of layers from 6 (baseline) to 4 improves validation loss from 1.7103 to 1.5268, even though the smaller model has a higher training loss (0.8501 vs. 0.6200), indicating stronger overfitting in the larger model (see [Table A.1.3](#)). More generally, the

relationship between model size and validation loss depends on the amount of available data. Larger models have higher capacity and can fit the training data more closely, but on small datasets this often leads to overfitting and worse validation performance. In contrast, with sufficiently large datasets, larger models can learn more complex patterns and achieve lower validation loss. In our experiments on Tiny Shakespeare, the dataset is relatively small, so increasing model size does not help and instead harms generalization. Therefore, bigger is not always better at this scale. This highlights the importance of matching model capacity to dataset size.

3. What role does dropout play? Compare the train/val loss gap across dropout values.

Dropout acts as a regularizer by randomly deactivating nodes during training, which forces it to learn more robust patterns instead of relying on specific features.

When we lowered dropout from 0.2 in the baseline to 0.1 in dropout01, the training loss dropped dramatically from 0.6200 to 0.2529, but the validation loss worsened sharply from 1.7103 to 2.2670 (see Table [A.1.3](#)). This produced by far the largest train–validation gap in all experiments, which is strong evidence that lower dropout caused severe overfitting.

This effect is also reflected in perplexity: while training perplexity improves ($1.86 \rightarrow 1.29$), validation perplexity degrades substantially ($5.53 \rightarrow 9.65$), confirming that the model memorizes the training data but fails to generalize.

4. Based on your experiments, what configuration would you recommend for this dataset, and why?

We would recommend **block128**. It achieves the lowest validation loss (1.4986) and lowest validation perplexity (4.48), indicating the best generalization among all experiments. In addition, it is also the fastest configuration (3.83 min), making it the best trade-off between efficiency and performance (see Table [A.1.3](#)). Compared to the baseline, block128 substantially improves validation loss ($1.7103 \rightarrow 1.4986$) while increasing training loss ($0.6200 \rightarrow 0.9449$), which reduces the train–validation gap. This indicates less overfitting and better generalization. Other configurations improve over the baseline but remain worse than block128. Lowering the learning rate ($1e-4$) achieves a similar validation loss (1.5090), but requires more than twice the training time. Reducing the number of layers (layer4) also improves validation loss (1.5268), but not to the same extent. Increasing the learning rate ($5e-3$) does not lead to further improvement (1.5421), and reducing dropout leads to much worse validation performance (2.2670), confirming the importance of regularization.

Overall, block128 provides the best trade-off between quality, generalization and efficiency among the tested configurations, and is therefore the most suitable choice for this dataset.

Conclusion

This study investigates the behavior of a small character-level Transformer language model trained on the Tiny Shakespeare dataset, with a focus on understanding how key hyperparameters affect optimization, generalization, and generated text quality. By systematically varying one parameter at a time relative to a fixed baseline, we were able to isolate and interpret the impact of learning rate, model depth, context length, and dropout regularization. The results demonstrate that, on small and highly structured datasets such as

Tiny Shakespeare, generalization is the primary challenge rather than optimization. While the baseline model achieved a relatively low training loss, it exhibited clear signs of overfitting, as reflected in a substantial gap between training and validation performance. This pattern became even more pronounced when regularization was reduced, confirming that small datasets require careful control of model capacity to avoid memorization. Across all experiments, reducing effective model capacity consistently improved validation performance. This effect was observed when lowering the learning rate, reducing the number of layers, and shortening the context window. Among all configurations, block128 performed best overall, as it achieved the lowest validation loss, the fastest training time, and produced the most coherent qualitative outputs. This suggests that limiting the context length not only improves computational efficiency but also acts as an effective constraint on the model, preventing it from overfitting to spurious long range patterns that are not reliably supported by the data. In contrast, increasing model flexibility without sufficient regularization proved detrimental. The reduced dropout configuration led to extremely low training loss but the worst validation performance, clearly illustrating the risks of overfitting. Similarly, increasing the learning rate introduced instability without yielding meaningful improvements, highlighting the importance of stable optimization dynamics. The qualitative analysis closely supports these findings. Models with lower validation loss produced more coherent and stylistically consistent text, maintaining dialogue structure and sentence flow more reliably. In contrast, overfitted or unstable configurations resulted in fragmented, repetitive, or syntactically inconsistent outputs. This confirms that validation loss is not only a numerical metric but also a strong indicator of perceived text quality. Overall, the experiments highlight a key principle in training language models at small scale, namely that model capacity must be carefully matched to dataset size. Larger or more flexible models do not automatically yield better results. Instead, controlled capacity and sufficient regularization are crucial for achieving good generalization. A limitation of this study is the relatively small dataset and limited training budget, which may restrict the performance of larger models. Future work could extend this analysis to larger and more diverse datasets where increased model capacity may become beneficial. Additionally, exploring longer training schedules or more advanced regularization techniques could provide further insights into scaling behavior.

In conclusion, this work demonstrates that even simple Transformer models exhibit complex interactions between hyperparameters, and that systematic experimentation is essential to understanding and optimizing their performance.

Bonus exercise

Bonus A: Language Modeling on Custom Data (WikiText-2)

For the bonus experiment, we trained the same character-level nanoGPT model on a custom dataset (WikiText-2) instead of Tiny Shakespeare. The goal was to analyze how strongly the training corpus influences the style, vocabulary, structure, and coherence of generated text. Since the assignment focuses on language modeling and text generation, using a stylistically distinct corpus provides a meaningful basis for comparison. The WikiText-2 dataset consists of verified Wikipedia articles and, at 13.3 MB, is substantially larger than Tiny Shakespeare. For preprocessing, the raw text files were concatenated into a single UTF-8 corpus. Punctuation, capitalization, and line breaks were preserved, as the model operates at the character level and therefore directly learns these surface patterns. No changes to tokenization were introduced, ensuring that the setup remained fully comparable to the baseline model. The model was trained using the same architecture and training configuration as the baseline. Quantitatively, the WikiText-2 model achieved a final training loss of 1.1242 and a validation loss of 1.2016 within approximately 8.05 minutes. The relatively small gap between training and validation loss indicates strong generalization, which can be attributed to the significantly larger and more diverse dataset compared to Tiny Shakespeare. This suggests that dataset size and diversity play a key role in stabilizing training and improving generalization. Qualitatively, the generated samples differ substantially from the Shakespeare baseline. While the baseline model produces text in an archaic, dialogue-driven style, the WikiText-2 model generates text in a neutral, encyclopedic tone (see Appendix B.2.1). It replaces archaic expressions such as “thou” and “hath” with modern vocabulary, including technical terms, named entities, and temporal references. A key observation is that the model learns not only linguistic patterns but also structural formatting. Similar to how the Shakespeare model reproduces speaker labels and dialogue structure, the WikiText-2 model captures Wikipedia-specific formatting conventions, such as section headers (e.g., “== History ==”) and parenthetical expressions. This demonstrates that the Transformer architecture, even at this small scale, is highly sensitive to the structural and visual characteristics of its training data. Despite strong local coherence, both models exhibit limitations in maintaining long-range consistency. However, the nature of their errors differs: the Shakespeare model generates stylistically consistent but semantically incoherent text, whereas the WikiText-2 model produces text that appears factually structured but may drift across topics over longer sequences. This reflects the higher linguistic diversity and complexity of Wikipedia data, which makes long-range coherence more challenging to maintain. In summary, this experiment highlights the strong dependence of language models on their training data. By changing only the dataset, the model transitions from generating stylized dramatic dialogue to producing structured, encyclopedic text. This demonstrates that character-level models are capable of capturing not only vocabulary and syntax, but also higher-level stylistic and structural properties of a corpus.

Bonus B: BPE Tokenization

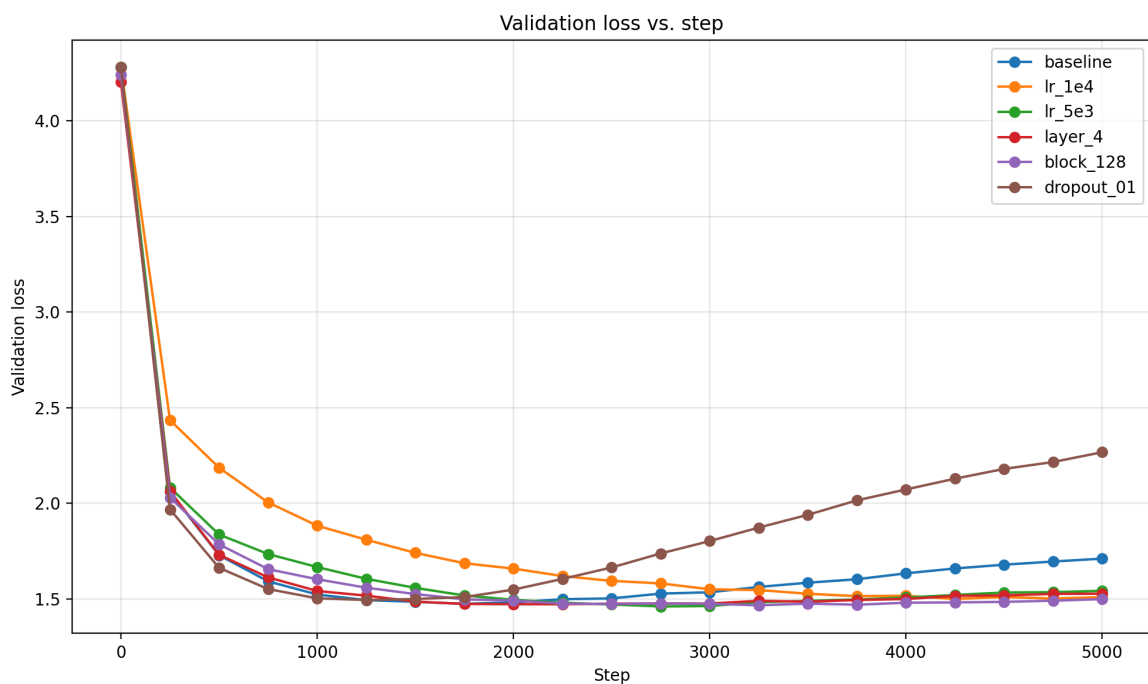
To evaluate the impact of tokenization on model performance, the standard character-level tokenizer (vocabulary size: 65) was replaced with a Byte Pair Encoding (BPE) tokenizer with a vocabulary size of 200. BPE is a sub-word tokenization algorithm that finds the most frequent pairs of adjacent characters in a text and merges them into new single tokens. This allows a model to represent common words as single units while still being able to build rare or unknown words out of smaller character pieces. By doing this, BPE balances the flexibility of character-level training with the efficiency of word-level processing. The goal of this experiment was to compare training efficiency, generalization performance, and qualitative output. On the Tiny Shakespeare dataset, the BPE model achieves a significantly lower training loss (0.2204) compared to the character-level baseline (0.6200), indicating that it can fit the training data more effectively. The training curves show a crossover at approximately 3500 iterations, after which the BPE model outperforms the baseline in terms of training loss. This suggests that BPE enables more efficient representation learning by encoding frequent character sequences as tokens. However, this improvement comes at the cost of generalization. The validation loss increases drastically to 4.443 (baseline: 1.7103), and the validation perplexity rises to 85.03 (baseline: 5.53), indicating severe overfitting. This demonstrates that while the BPE model is highly effective at memorizing training patterns, it fails to generalize to unseen data. One likely explanation is that the BPE vocabulary becomes overly specialized for the relatively small Tiny Shakespeare dataset. By merging frequent character sequences into tokens, the model may overfit to specific patterns in the training data rather than learning more generalizable linguistic structures. In contrast, character-level tokenization enforces a more fine-grained representation, which acts as an implicit regularizer. In terms of efficiency, training time remains nearly identical (~7.9 minutes). While BPE reduces sequence length and allows the model to process longer contexts per step, this advantage is offset by the increased complexity of the larger vocabulary. To further isolate the effect of tokenization, BPE was also applied to the larger WikiText-2 dataset. In contrast to the results on Tiny Shakespeare, the negative generalization effects of BPE were less pronounced on this dataset. Due to the significantly larger size and higher diversity of WikiText-2, the learned token merges are more broadly applicable and less likely to overfit to specific sequences. This highlights an important interaction between tokenization and dataset size: while BPE can lead to severe overfitting on small datasets, it becomes more effective as dataset size and diversity increase. Larger corpora provide sufficient variability for subword units to generalize across contexts, allowing BPE to better capture meaningful linguistic structure. This suggests that tokenization interacts strongly with dataset size and should not be considered independently. Overall, the experiment shows that increasing tokenization granularity can significantly improve training efficiency but may harm generalization when the dataset is small. For Tiny Shakespeare, character-level tokenization provides a better balance between model capacity and generalization, whereas BPE with a vocabulary size of 200 leads to excessive memorization. In contrast, on larger datasets such as WikiText-2, BPE becomes more suitable due to improved generalization across a broader range of text patterns.

Appendix

Appendix A: Main Task Experiments (Baseline & Hyperparameters)

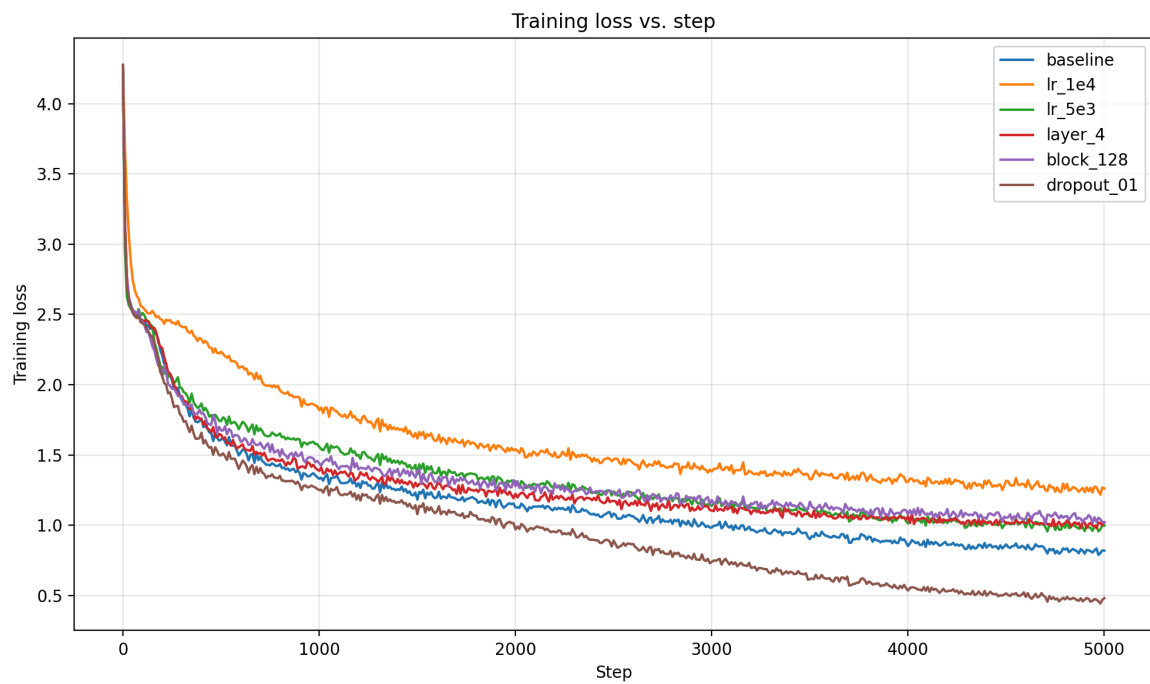
A.1 Performance Plots:

Plot A.1.1: Validation loss



The graph shows the validation loss of the baseline algorithm and experiment runs with single changed parameters over 5000 iterations

Plot A.1.2: Training loss



The graph shows the training loss of the baseline algorithm and experiment runs with single changed parameters over 5000 iterations

Table A.1.3: Summary results

Experiment	LR	Layers	Embd	Block	Dropout	Iters	Train Loss	Val Loss	Train Perplexity	Val Perplexity	Time (min)	Out dir
block_128	0.001	6	384	128	0.2	5000	0.9449	1.4986	2.57255611	4.475419096	3.83205818	out-shakespeare-block128
lr_1e4	0.0001	6	384	256	0.2	5000	1.1538	1.509	3.170216874	4.522206326	7.908074693	out-shakespeare-lr1e4
layer_4	0.001	4	384	256	0.2	5000	0.8501	1.5268	2.339880828	4.603422276	5.310454783	out-shakespeare-layer4
lr_5e3	0.005	6	384	256	0.2	5000	0.852	1.5421	2.344330828	4.674396203	7.895879218	out-shakespeare-lr5e3
baseline	0.001	6	384	256	0.2	5000	0.62	1.7103	1.858928042	5.530620415	7.922770537	out-shakespeare-baseline
dropout_01	0.001	6	384	256	0.1	5000	0.2529	2.267	1.287754495	9.650406125	7.879188657	out-shakespeare-dropout01

The table compares the architecture of the algorithms from the different experiment setups and the outcomes after 5000 iterations

A.3 Model Samples:

Below are the three generated samples of 500 tokens for each experiment: baseline, block_128, dropout01, layer4, lr1e4 and lr5e3.

Sample Output A.3.1: Baseline

```
Overriding: out_dir = out-shakespeare-baseline
Overriding: device = cuda
Overriding: num_samples = 3
Overriding: max_new_tokens = 500
Overriding: seed = 1337
number of parameters: 10.65M
Loading meta from data/shakespeare_char/meta.pkl...
```

Clown:

So, who is he wounds himself aboard and his takes
the hands are the purpose of baredilation, and they are starried out
about me for the court. Think you might so, mine acts that
violent in me now of your spelling and unto your children; your
solemn years make no more so visitation.

CORIOLANUS:

Nay, so what evil you are done?

Third Citizen:

And she is
to known breathe of this. Why, there is not the people?

MENENIUS:

I see your brother daughters, he must not to say tell you
show your voic

Menenius, I must tell you have me one and seem another
weak thrown them against the law; but, with
confessions to castly. A biass of alliance, a bawd,
being very dispatch'd within me; throw is it
doubtle fellow, any better of a poor sound of a
very of gates and bear-so grave-buried a soul. Do you not, good flatter
this and the courts of the cause of a son, Camillo.

LUCIO:

Come, stay not the submission; and wilt the down be almost
like a sister of the prisoner. Speak thou the glory fly? I will b

MARIANA:

I beseech you, I would where you can be, here in the face
That the shepherd have sweat and so letters her first:
I have done would not a blow of Edward and his hands;
But the sevent and do this blame at my breast.

KING EDWARD IV:

Cousin, I will be done; your Hastings for love,
Have made the fouless of Lancaster Henry,
Show'd my brother shall die.

Provost:

For thine own son England's coverby,
That Edward shall be holder'd his brow to my wrong,
And deliver'd in our Edward's royal king,

Sample Output A.3.2: block128 (context length block size 128, 4 heads)

```
Overriding: out_dir = out-shakespeare-block128
Overriding: device = cuda
Overriding: num_samples = 3
Overriding: max_new_tokens = 500
Overriding: seed = 1337
number of parameters: 10.65M
Loading meta from data/shakespeare_char/meta.pkl...
```

KING RICHARD II:

Say that stands you both to take and me:
So takes a thing I will be thy determine,
And that I fall, an old peace, a toad,
I can tell this bound is ensuing with the cloth to a
death that will answer him to ease the maid with a creative
how you love upon my execution.

DUKE VINCENTIO:

Nay, so we will be an enemy's wife, I have stood
For his gracious nurse, I would not say 'tis no fled
at the darken of our rest, being behold a posterns
Of his brother to a gate
His promise and vict

Men pardon, mark me with the earth; and I'll meet my
branches ready that I live on thee; but, whom
I have made thee so, my man should perjure
Thee at Antium alteried as the soldiers of my
sovereign as the enemy, I shall be so.

First Senator:

How, monstrous!

CORIOLANUS:

How brings me! what's that?

CORIOLANUS:

It cannot be.

CORIOLANUS:

I know the noise.

CORIOLANUS:

Who is it?

LEONTES:

I am the matter, on what thou art a glassion of kings
To have the power where thou wouldst not fly,
For thou

Men that I both. Their cursed king.

LADY GREY:

Come, my lord; I would you be well, sir, we say,
Ye see them his body to be the house of York,
And with a valour and his careful traitors,

And all in this blame at my breasts dissonmenting.

DUKE OF AUMERLE:
The time were something the land with thousand soldiers
Would not murder, and but breath it is contrary.

QUEEN MARGARET:
Ay, my sovereign, a gross steel, to be the base.

KING RICHARD III:
Good friends and grave pardon you of this rage,
That t

Sample Outputs A.3.3: dropout01 (regularization dropout 0.1)

Overriding: out_dir = out-shakespeare-dropout01
Overriding: device = cuda
Overriding: num_samples = 3
Overriding: max_new_tokens = 500
Overriding: seed = 1337
number of parameters: 10.65M
Loading meta from data/shakespeare_char/meta.pkl...

All:
If this infant is not to be so, but being a
great thing talks: 'tis a put of his barket--
was, away, what follows himself of the office
Becomes his little birdens, and that Henry or the deed.

BENVOLIO:
No, sir, indeed, but she comes your graces all,
you love upon my attorneys.

BRUTUS:
Not a letter of what evides to the people
Than it compounteous; the exervices that we are
To the fire-paral of the people?

MENENIUS:
I shall took it:
Sadnable it graces of the preyest your husband's form;

Men part, sir, you shall have hansorous most of young.

LUCIO:
Was not the city of the good world,
confesses should sulprent instrument it,
But weary when moved their successits from my bosom:
Summers that Richard yield Claudio shows my son,
And dear mother than an envy were but wrath.

DUKE VINCENTIO:
What lack is this, man?

ISABELLA:
O, ay, poor noisom, Camillo.

LUCIO:

Sir, disposition me not, I prate your bed:
I came the senators be to your humour.

ISABELLA:
Well, would have all this good

Men take it well against the common with the coronation.

KING EDWARD IV:
What art thou, we say, that fled me infirmity to his like a word,
And all the good sword I have a cause of that is.

BAGOT:
Be gone more than this excelliw made me have eat of mine
Than thou wert the spamp of the spoil, thou art of just.
Where is Master Oxford?

Nurse:
I think there is thy dear hold on the harsh.

Nurse:
What blood is this lamb a friar?

ROMEO:
Banish'd for where?

Nurse:
No, I wish, I am no warrant, thou

Sample Output A.3.4: layer4 (model depth 4 layers)

```
Overriding: out_dir = out-shakespeare-layer4
Overriding: device = cuda
Overriding: num_samples = 3
Overriding: max_new_tokens = 500
Overriding: seed = 1337
number of parameters: 7.11M
Loading meta from data/shakespeare_char/meta.pkl...
```

Which if this my brother's blood is to Baynard's shame,
My hand of barnet heavy company?
Where's the sweet prince's noble knee
Your proceeding brother Gloucester is enter'd;
Once Henry of Edward's knight of York,
Welcome, tell the Duke of York and Sir Welshmen:
My Lord of York and Derby, I thought keep Willongbroke.

KING EDWARD IV:
Then, in thy sovereign sense and noble trust
Are my tongue to piece Edward's brother's son:
Thou art too fight and betwixt his brothers.

WESTMORELAND:
Who cannot be

Men part, I must tell you gave me
That I have spoken your mind. This worst can we not be born.

First Servingman:

What, man?

MENENIUS:

Sir, I have desire moved the people of this?

Second Servingman:

The gods are you, good my lord, the violent news,
Content the gates of your faces: the least of your death well
In this and the city, if I shall possess
To do this man to do it.

Conspirator:

O thou art my father, and good for all.

BENVOLIO:

The duke is proud to a choler, would have a goon well.

Men than I both. This is the new-gentleman
And had been disadvised to action him:
But we were mine eye to worshipful duke,
His mistress shadows a blood and bride
AIntasted their city, and be a common of blame,
When break singular magistrate sues
I will not speak a fool, and famous love,
How the old consort and so soon arm as he is,
As I should be his; and therefore, he that I be heard
Would war it not see to give the babes hour.

Second Keeper:

Ay, if they be gone, but so, I am contented
To see

Sample Outputs A.3.5: lr1e4 (learning rate 1e-4)

```
Overriding: out_dir = out-shakespeare-lr1e4
Overriding: device = cuda
Overriding: num_samples = 3
Overriding: max_new_tokens = 500
Overriding: seed = 1337
number of parameters: 10.65M
Loading meta from data/shakespeare_char/meta.pkl...
```

Nurse:

Redicle, and then, being it to Barnard's shrives;
For takes and their will be traitors,
Have away with himself though thou speak thy head,
Which in thy son weeds accessation,
Thou that thou long'st believed like death.

HASTINGS:

Ah, at princess, why most not know thee gods
More like to thee, or I show thy city,
To thou shalt conscience of thy life.

QUEEN MARGARET:

I'll speak when thou hast on my sovereign,
So murder subjects by the advance of me,

You are my cause no much stoop my count

Menening, and good such on a woman;
 Shall seem poor between throat straight in this life.

WARWICK:
 There's the people, many is so the earth.

LADY ANNE:
 How! there is state our father's times to death.

FiRst Servingman:
 I have now the vile head, and there could a world
 Have the way so leving manner, therefore I must dead?

GLOUCESTER:
 Nay, but I am in the eaten of the other wedd.

CLARENCE:
 It is approachess be that was much person.

CLARENCE:
 My lord, because I love, would I fear;
 To know you

Men that I beseech your husband;
 And her the very heart disatch'd of your horses!

GLOUCESTER:
 And let me were not, in this ight care would
 Do call the good sword. In thy are the cursed
 Of changed world, blame at my breathes
 And witness there gentleman, It did thou wert heart,
 And makes our presence consuls be defend,
 So much have lived their sits hands well becomes,
 For they are have been some caventure.

GREGORY:
 Not believe me; we would upon my wife tender to light.

WARWICK:
 Richard, my lord,

Sample Outputs A.3.6: lr5e3 (learning rate 5e-3)

```
Overriding: out_dir = out-shakespeare-lr5e3
Overriding: device = cuda
Overriding: num_samples = 3
Overriding: max_new_tokens = 500
Overriding: seed = 1337
number of parameters: 10.65M
Loading meta from data/shakespeare_char/meta.pkl...
```

All thy brows will be long to be set boaster:
 What thy sweet lands are thy purpose to be determine,
 And thy heart thou hast pray'dst to thy heart and my life
 When tender contempt my drop to thee to thee:
 And masterly in a wallower thou hast prepared;
 Or why have I seen my father should like to thee

And in thy dear in the earth's world eyes?
And so much for the king thrust for the rise of English leave,
That Prince Edward's house! still I am advanced
With thy first to my son
Hath to change the w

Men pardon me with such a game.
Is not thyself and fortune's madam? what say you the world
Is no more fresh than that loves till we lose.

SLY:
See, my lord.

Lord:
Call Lord:
And me be so;
On the haste of Richard, the Gaunt; and the queen heart.
For me to have been to give them with some victory
And death will I send yours.

Lord:
All I shall post to him.
Call your grace is in our own secrets!
But is your cousin of your cousin English king!

Lord Marshal:
My wife, therefore is not foul in gifte

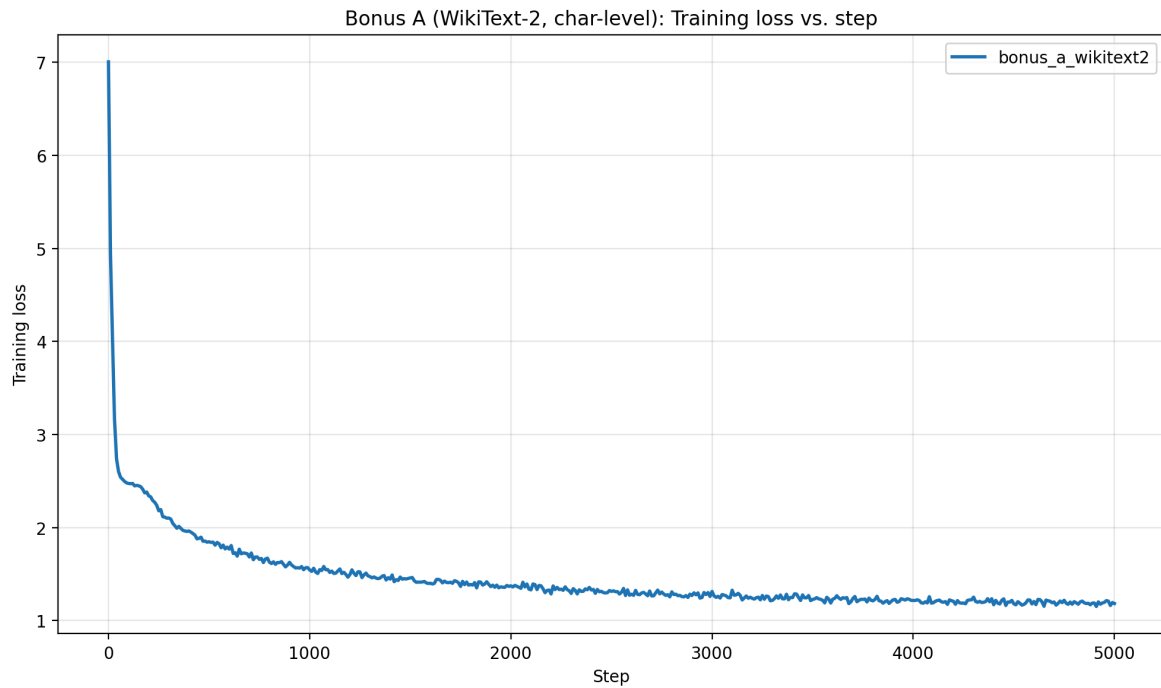
Messenger:
The power, I'll stay; go with me all,
I cannot say your grace.

LEONTES:
My lord,
Am I seek to were not the storm of his face
Than he should go abhorse. I have a cause
Of all times and my perform me at my breasts
Should make my happy parting and descent.
But He was too lately spoke,
Because to me as he stoops thus have done,
Nor by his shock disposed in his holy lives;
But his daughter is but found to my victory.
On his heart the honour will tell me what thou so ill
My son I will have

Appendix B: Bonus Exercise A (Wikitext2 Analysis)

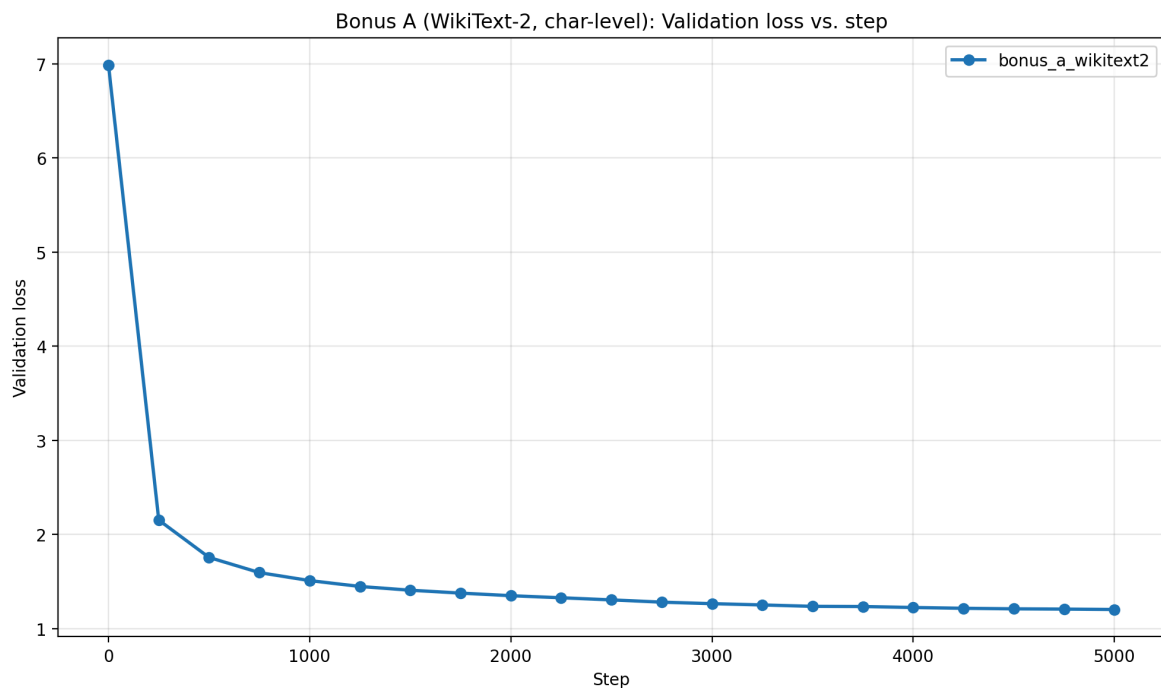
B.1 Performance Plots:

Plot B.1.1 Bonus A: Character-level nanoGPT model on WikiText-2 training loss



The graph shows the training loss using the custom dataset WikiText-2 over 5000 iterations

Plot B.1.2 Bonus A: Character-level nanoGPT model on WikiText-2 validation loss



The graph shows the validation loss using the custom dataset WikiText-2 over 5000 iterations

B.2 Comparative Data:

Table B.2.1 Bonus A: Character-level nanoGPT model vs custom BPE comparison

Run	Train Loss	Val Loss	Time (min)
bonus_a_custom_char	1.1242	1.2016	8.05126285
bonus_b_custom_bpe_wikitext2_bpe_bonuss_200	1.9236	2.1656	8.076443035

The table compares the outcome for the character level nanoGPT model and the Byte Pair Encoding tokenizer over the custom dataset WikiText-2 after 5000 iterations

B.3 Sample Outputs:

Sample Outputs B.3.1 Bonus A: WikiText-2 sample outputs

```
Overriding: out_dir = out-wikitext2-bonus
Overriding: device = cuda
Overriding: num_samples = 3
Overriding: max_new_tokens = 500
Overriding: seed = 1337
number of parameters: 11.05M
Loading meta from data/wikitext2_char_bonus/meta.pkl...
```

= = Post @-@ combat and the early Post @-@ Commando Roose = =

A lesson of the 2010 World Cup was announced that the city involved in the command of the German legislature in 1936 . Germany signed in that in 2015 , and on she was the started post of the starts of a first season of a first number on the Confederacy for the season .

In 1970 , the start of Pearl Start , a Canadian album , a First Simpsonship Confederate action debuted in the Billboard Hot 10 episode , and White Hot 100 had

Serenants have been suggested that the British army came to have a " rate " the close @-@ defense " of his father at Boston Plat , the comedy was given to build for high @-@ defense pieces ; he also came to win a total lawsuit , and showed his east .

= = = Fifth century = = =

At the same day , Henry Saram showed his partnership with the Sinai near the Caribbean defenders he was unable to the part of the same time , but was the fifth appearance at Caribbean Distance State , but she wa

= = = Great summer and contract = = =

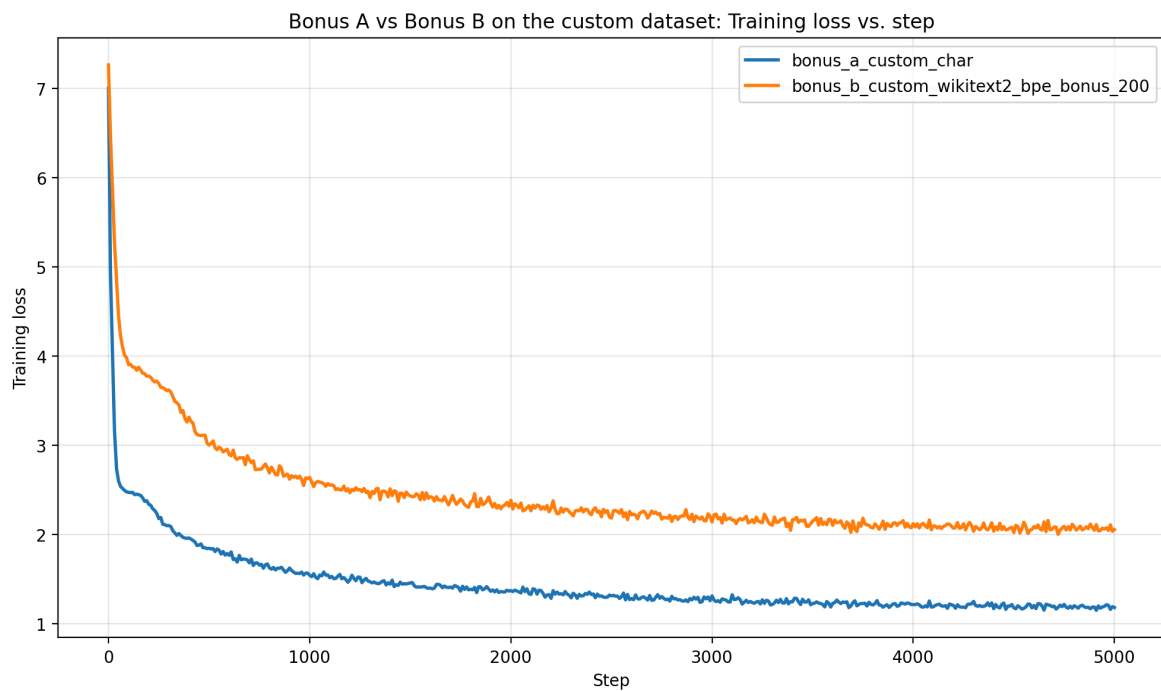
Creative authorities were responsible for the country . The early controversy and British areas , where the contemporary successful commandment was published . The team was achieved into the English Summer Olympics and the Maryland Maryland . Later , the British area had already never been rejected in a physical career , with the game 's fatality tried to the neighborhood , which is built in September 2007 .

In 2000 , the British marriage was dec

Appendix C: Bonus Exercise B (BPE & Shakespeare)

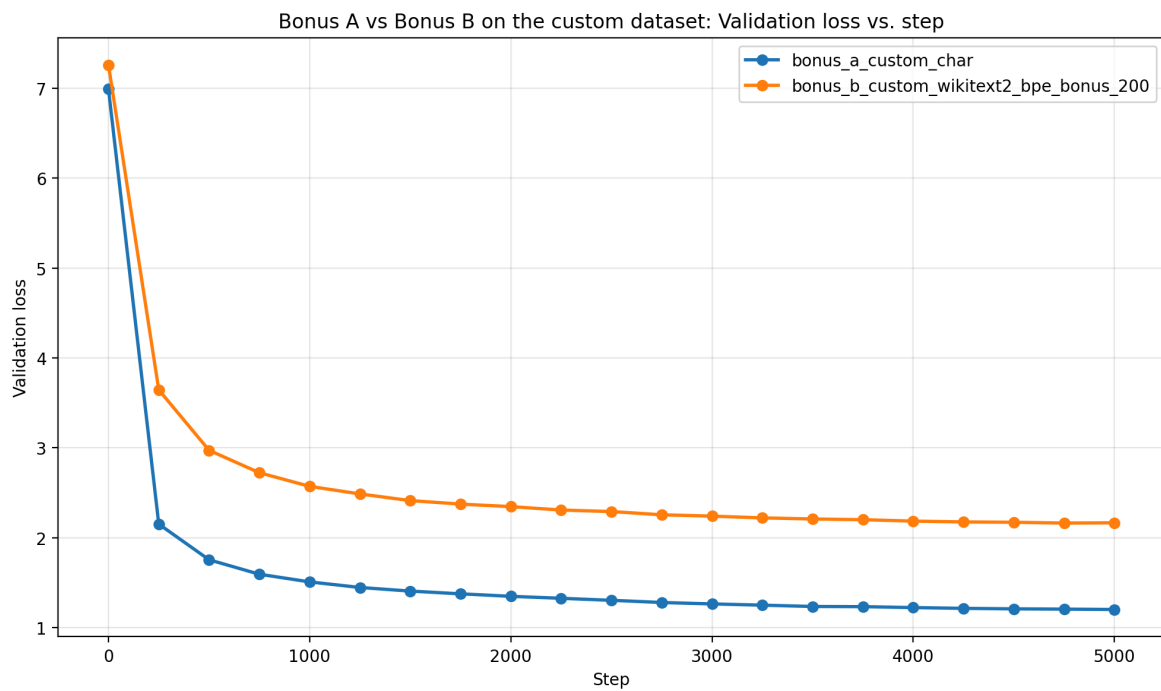
C.1 Performance Plots:

Plot C.1.1 Bonus B: Character-level nanoGPT model vs custom BPE training loss



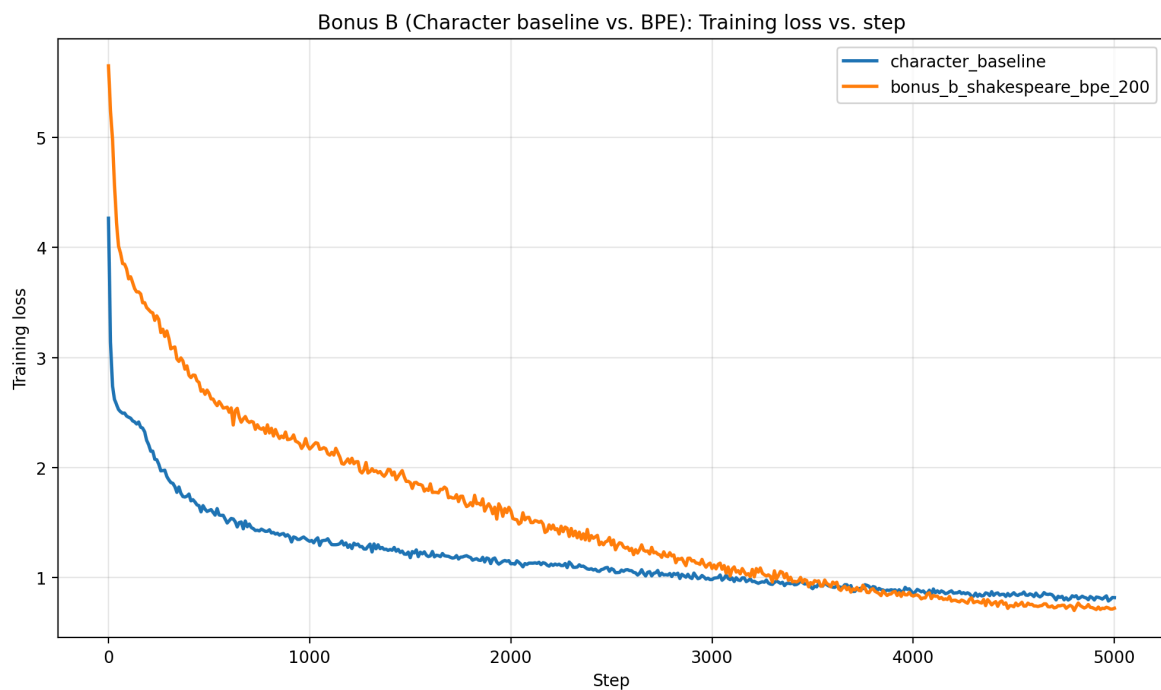
The graph shows the training loss from Bonus A in comparison to the BPE from Bonus B using the custom dataset WikiText-2 over 5000 iterations

Plot C.1.2 Bonus B: Character-level nanoGPT model vs custom BPE validation loss



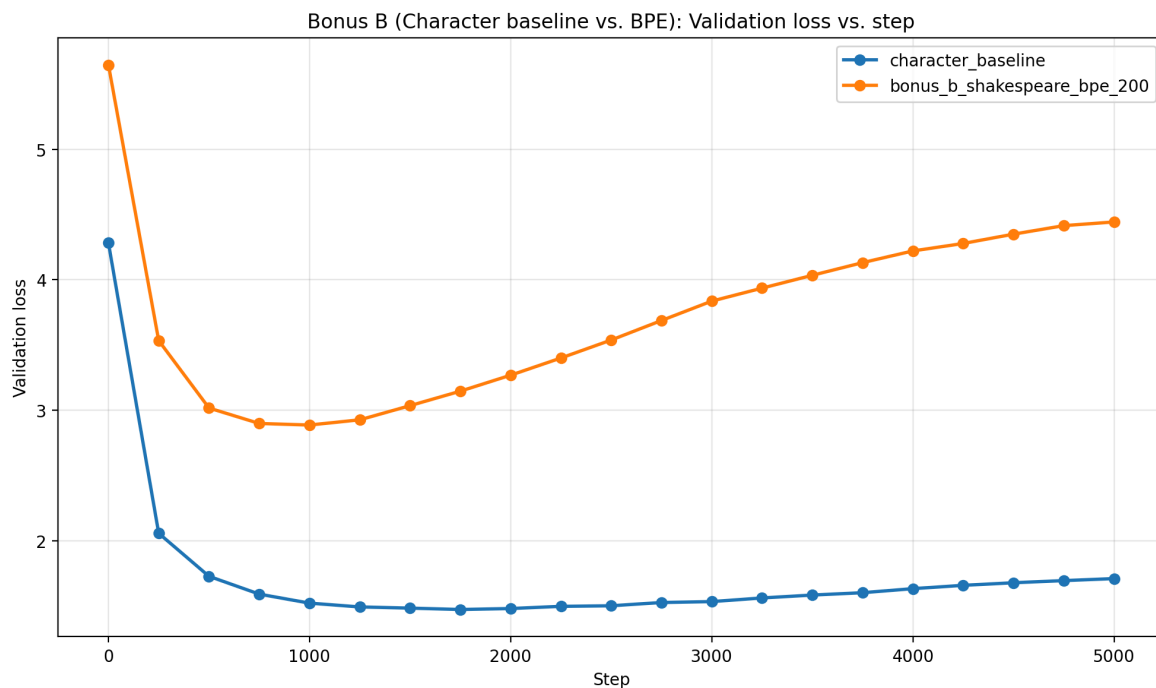
The graph shows the validation loss from Bonus A in comparison to the BPE from Bonus B using the custom dataset WikiText-2 over 5000 iterations

Plot C.1.3 Bonus B: Baseline vs BPE comparison training loss



The graph shows the training loss from BPE from Bonus B in comparison to the baseline using the custom dataset WikiText-2 over 5000 iterations

Plot C.1.4 Bonus B: Baseline vs BPE comparison validation loss



The graph shows the validation loss from BPE from Bonus B in comparison to the baseline using the custom dataset WikiText-2 over 5000 iterations

C.2 Comparative Data:

C.2.1 Bonus B: Table comparison of baseline vs BPE

Run	Train Loss	Val Loss	Train Perplexity	Val Perplexity	Time (min)
character_baseline	0.62	1.7103	1.858928042	5.530620415	7.922770537
bpe_shakespeare_bpe_200	0.2204	4.443	1.246575261	85.02964837	7.876097303

The table compares the outcome for the baseline model and the Byte Pair Encoding tokenizer using the Tiny Shakespeare dataset after 5000 iterations

C.3 Sample Outputs:

Sample Outputs C.3.1 Bonus B: Shakespeare BPE 200

```
Overriding: out_dir = out-shakespeare_bpe_200
Overriding: device = cuda
Overriding: num_samples = 3
Overriding: max_new_tokens = 500
Overriding: seed = 1337
number of parameters: 10.72M
Loading meta from data/shakespeare_bpe_200/meta.pkl...

That's in their own.
```

POLIXENES:
Have you both even hen
Should have knowledged of heart.

FLORIZEL:
No, by great kindly,
Go joyful it of;
Lay I pray, I pray thee, go curse
For some remembering: the nobles shall buy, Warwick,
Camillo believe them and a foe.

Second Lord:
We're well, sir?

First Servant:
We'll three on any thing for, if you find
By those field me under youth,
I shall not try them.

PERDITA:
You cannot not away:
I say, sir.

PERDITA:
Ay, fellow!
Then let them not well.

FLORIZEL:
Now will meet well.

POLIXENES:
Come, come, madam.

Servant:
Bohemia, what news?

First Servant:
I have been to-night.

LEONTES:
I will buy it.

First Servant:
Ay, by
Will I had rather under the city's death,
And that I have no sent to him.

PAULINA:
Sir, nor I have hear, sir,
Nor nor one to tite of thine.

PAULINA:
Master, condemn'd big it. You have heards other
To pluck us of my self: but 'tis no letter,
Even for some

Of your grace's brother's son.

GLOUCESTER:
Welcome, sir, I pray thee, Catesby; 'tis alasy.

BUSHY:
My lord, I like my poor lord.

KING RICHARD II:

The earldom of the king is climity.

BUCKINGHAM:

What is the good cousin of the prince?

KING RICHARD II:

Why, what is the time was the matter?

BUCKINGHAM:

So presently to take the castle?

BUCKINGHAM:

When it is the champister.

KING RICHARD II:

Ay, as we are, when they stay at our chambery.

NORFOLK:

For they at the crown?

BUCKINGHAM:

Ay, my lord.

KING RICHARD III:

Say we shall.

BUSHY:

My lord.

KING RICHARD III:

Sweet poliath, and I had--

Would it were cried in received.

My lord, I have resolute days as my love?

GLOUCESTER:

Why, so he was the king comes of sovereign.

BUCKINGHAM:

Here comes the crown and his conscience!

CATESBY:

Richard Gloucester, he that hath all.

BUCKINGHAM:

God and Hastings, and your queen,

I had said leave met

A bottom of yourself side, I know the trongue

And sent my prince; methinks it not: as I have a

glass here over love. If you do content

not no hope, this shall be your brother

And proceedings his rest pernicious blandness; and to be kindly

heard to do me buy yourself and yourself.

LEONTES:

O, I met thee

To keep this needful:

Massay it is ever, I could flatter, I hope

His castless done courtesy; and if am pass'd flock; or no fles

If he would have all the setts of the way,

I'll tell it what he would put myself.

LEONTES:

Two will be

Her depart.

LEONTES:
Prayer, I have heard you then, yet how he did
When so little too?

ANTIGONUS:
I dare not.

LEONTES:
Not at it.

LEONTES:
I am for these large, sir.

PAULINA:
You had been no offences to see what's of
When they would burnt themselves.

LEONTES:
I beseech you, good night,
Who persuade the air of the trembles:
Therein I thought them can be;
Whom thou speak'st, I a

Sample Outputs C.3.2 Bonus B: WikiText-2 BPE 200

```
Overriding: out_dir = out-shakespeare_bpe_200
Overriding: device = cuda
Overriding: num_samples = 3
Overriding: max_new_tokens = 500
Overriding: seed = 1337
number of parameters: 10.72M
Loading meta from data/shakespeare_bpe_200/meta.pkl...
```

That's in their own.

POLIXENES:
Have you both even hen
Should have knowledged of heart.

FLORIZEL:
No, by great kindly,
Go joyful it of;
Lay I pray, I pray thee, go curse
For some remembering: the nobles shall buy, Warwick,
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BUCKINGHAM:
So presently to take the castle?

BUCKINGHAM:
When it is the champister.

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Ay, as we are, when they stay at our chambery.

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For they at the crown?

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I had said leave met

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And sent my prince; methinks it not: as I have a
glass here over love. If you do content
not no hope, this shall be your brother
And proceedings his rest pernicious blandness; and to be kindly
heard to do me buy yourself and yourself.

LEONTES:
O, I met thee
To keep this needful:
Massay it is ever, I could flatter, I hope
His castless done courtesy; and if am pass'd flock; or no fles
If he would have all the setts of the way,
I'll tell it what he would put myself.

LEONTES:
Two will be
Her depart.

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Prayer, I have heard you then, yet how he did
When so little too?

ANTIGONUS:
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When they would burnt themselves.

LEONTES:
I beseech you, good night,
Who persuade the air of the trembles:
Therein I thought them can be;
Whom thou speak'st, I a

Appendix D: Log Files

The log files for all experiments, including the bonus tasks, are available at the following Google Drive link:

https://drive.google.com/drive/folders/1uS2VMs9J1Ev0S8g-ehGuA3M1VxzMvJBB?usp=share_link